Full-Waveform Classification and Segmentation-Based Signal Detection of Single-Wavelength Bathymetric LiDAR

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***Abstract*— Single-wavelength bathymetric light detection and ranging (LiDAR) (532 nm) can provide seamless meter- and submeter-scale digital elevation model (DEMs) of both the terrestrial surface and seafloor. However, mixed terrestrial and bathymetric surfaces obtained by this sensor are challenging for full-waveform (FW) signal detection. This study addresses the issues in two FW mixed surfaces: accurate classification of terrestrial and nonterrestrial waveforms from the original waveforms without auxiliary information and flexible detection of peaks based on a new FW theoretical model. A novel FW signal detection model (FWSD) for single-wavelength bathymetric LiDAR is proposed without complex feature extraction and iterative procedure through waveform classification and segmentation. The raw FWs are divided into five categories for subsequent signal detection using a convolutional neural network that merges local descriptors with contextual information. The signal detection task is then split into FW segment recognition and peak extraction using a new FW model, which integrates a leapfrog sliding window FW segmentation, an improved extreme learning machine (ELM) algorithm for FW segment recognition, and a flexible signal detection framework. To search for the optimal initial parameters for ELM, a self-annealing particle swarm optimization (SAPSO) algorithm is introduced, and the output weight is adjusted by online sequence to improve its generalization. When combined with the Richardson–Lucy deconvolution (RLD) algorithm, FWSD can be adapted to deal with shallow water waveforms. Finally, a test demonstration with an airborne dataset shows that FWSD has higher detection efficiency and higher accuracy than Levenberg–Marquardt algorithm optimized generalized Gaussian model (LM-GGM) and RLD algorithm.**

***Index Terms*— Airborne bathymetric LiDAR, full-waveform (FW), signal detection, single-wavelength.**

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I. Introduction

A. Background

COASTAL management and monitoring are indispensable tasks faced by policy makers and scientists globally. However, it is still a challenge to obtain high-accuracy and high-density data with high rates of recovery efficiency in coastal zone. Bathymetric data in the range of more than 10 m are derived mainly using echo sounders (single beam and multibeam) [1], [2]. For unnavigable areas, the traditional surveying methods, dependent on a real-time kinematic global navigation satellite system, total station, or unmanned ships, will be commonly used, even though these methods are generally considered as time-consuming and labor-intensive approaches [3], [4]. Although satellite altimetry and satellite-derived bathymetry can generate large-area medium-resolution digital elevation model (DEM) data at low cost, poor universality, strong dependence on ground-truth data, and low inversion depth accuracy have limited its application as a main supplementary means in coastal underwater topographic mapping [5], [6].

As an active remote sensing technique characterized by high resolution, low cost, and high flexibility, airborne laser bathymetry (ALB) has great potential to satisfy the urgent demands and fill out the gap mentioned above in terms of coastal mapping [7]. The traditional dual-band ALB systems are usually equipped with both near-infrared and green lasers, for example, Optech SHOALS-1000/3000, CZMIL (Nova), Titan, and Leica Hawk-Eye II/III [8], [9], [10]. Compared with the dual-band ALB system, due to its low cost, portability, and applicability, the single-wavelength ALB system, such as the NASA EAARL, Fugro LADS Mk3, REIGL VQ-820G, Fibertek CATS, and NCALM Aquarius, which is only operated with a single sensor, proposes an ideal selection for shallow-water bathymetric mapping [11], [12], [13]. Generally, the energy of a green laser pulse is not enough to penetrate the depth of over 15 m [14]. However, under the same flight conditions, the underwater laser point density obtained by a single-band ALB system can be at least several times orders greater than that acquired using a dual-band system.

B. Related Works

1) Peak Detection: In a small-footprint full-waveform (FW) LiDAR (FWL) system, the FW records the entire digitized

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backscatter laser pulse received by a detector with high sampling rate [15]. Using a single-wavelength bathymetric LiDAR system, which operates with a single independent shot, to determine the portion of the light trajectory through water is an arduous task. FW signal detection is one of the foundation and key points of single-wavelength ALB data processing, and the corresponding methods can be classified into the following three categories: 1) echo detection, which focuses on detecting echo locations without considering the radiometric characteristics of the target [16], [17]; 2) mathematical approximation, using a single mathematical function or a combination of multiple functions to describe FW [18], [19]; and 3) deconvolution, which recovers the target response by removing the emitted component from the returned waveform [20], [21], [22]. For topographic LiDAR, the Gaussian decomposition approach can be appropriately used in most applications [23]; that is, statistically, up to 98% of waveforms can be fit by a Gaussian function [24]. Certain Gaussian decomposition methods have been transferred to bathymetric waveform processing [25]. According to the ALB theory, a green waveform is described as linear superposition of the surface return, volume backscatter return, bottom return, and noise [26]. Hence, it is inappropriate to use the Gaussian decomposition method in bathymetric waveform signal detection as a consequence of the water column return is on longer suitable for fitting with a Gaussian function anymore [27]. Subsequently, surface–volume–bottom decomposition methods are presented to adapt to the bathymetric waveform, and specific functions are used to describe the water volume echo, such as triangular [19], quadrilateral [28], exponential, and multiple exponential segments [29]. The Richardson–Lucy deconvolution (RLD) algorithm has been proven to perform well in bathymetric waveforms with a very shallow depth and a weak bottom response [30]. However, most related previous work that is focused on rigorous mathematical models requires time-consuming and complex calculations. Moreover, no single method could fit all the waveforms.

The mixed terrestrial and bathymetric surfaces pose an extra challenge to waveform processing, especially when it comes to shallow water. It indicates that straightforward waveform-by-waveform processing cannot correctly abstract seafloor topography data from waveforms. However, due to reasons such as the sea surface refraction phenomenon and different propagation media, terrestrial, and nonterrestrial signals corresponding to different coordinate solution geometry model, it is of particular necessity to identify FW categories. So far, lack of appropriate methods, most of the signal detection models do not support or take it into consideration.

2) Waveform Classification: Numerous ALB land–water classification studies have received much attention which can be classified into the following three categories: Raman band (647 nm) equipped [31]; aerial photogrammetry, satellite remote sensing imagery, or video-assisted (Parker and Sinclair, 2012); and waveform-feature-based [32] methods. However, most mainstream commercial bathymetric LiDAR systems (except SHOAL-1000T and SHOAL-3000, etc.) are not equipped with Raman band owing to excessive hardware costs. Moreover, although numerous devices integrate auxiliary sensors (e.g., multispectral or hyperspectral sensors), which enable the devices to collect point cloud data and remote sensing images simultaneously [33], the auxiliary data are generally considered complementary rather than essential. The waveform-feature-based approaches, such as the morphological characteristics of different lasers [34], convolutional features [35], local descriptors with context information [36], multichannel FW parameters [37], [38], and threshold intervals of water surface points [39], can directly perform waveform classification without the assistance of auxiliary data. Although numerous LiDAR point cloud classification approaches for topographic LiDAR mapping systems [40], [41] and multichannel bathymetric LiDAR exist, classification of single-wavelength waveforms remain few in number. The classification of single-wavelength waveforms mainly depends on the waveforms' characteristic extraction (e.g., amplitude, projected area, and width), which has close ties with waveform preprocessing and decomposition. Thus, the computation load increases by at least one order when compared with other methods.

As mentioned above, the approaches' selection of appropriate waveform classification and decomposition are of vital importance for guaranteeing high accuracy and fast response. Nowadays, machine learning has become increasingly popular and widely promoted, and it has demonstrated excellent performance in various fields such as image, speech, and natural language processing. The objective of this article is to construct a new signal detection model FW signal detection (FWSD) for single-wavelength ALB system based on the proposed FW theoretical model. Furthermore, to avoid complex calculation, machine learning methods are induced for intuitive and efficient signal detection, in which a deep learning framework is constructed to automatically divide the ALB waveforms into land, water, and mixed interference waveforms, and a self-annealing particle swarm optimization (SAPSO) online sequential extreme learning machine (OS-ELM) method is used to perform waveform segment classification. Finally, the maximum peak (MP) and RLD algorithm [42] are combined in a flexible framework for peak extraction under the constraints of waveform labels, which can be adapted to different waveforms and different water situations with high flexibility and high accuracy.

II. Methods

A. FWSD Model Structure

The approach exploits the human cognitive process on 1-D signal processing to avoid intricated signal preprocessing and parameter determination. Deriving from this, it is extended to process single-wavelength ALB waveforms, and a novel FW signal detection (FWSD) model is proposed. The schematic of the FWSD model processing flow can be summarized briefly as waveform classification and peak detection, as illustrated in Fig. 1.

1) Waveform Classification: Each FW will be recognized as one of the five categories, based on the constructed deep convolutional neural network (CNN), in which invalid waveforms will not be involved in the following operations.

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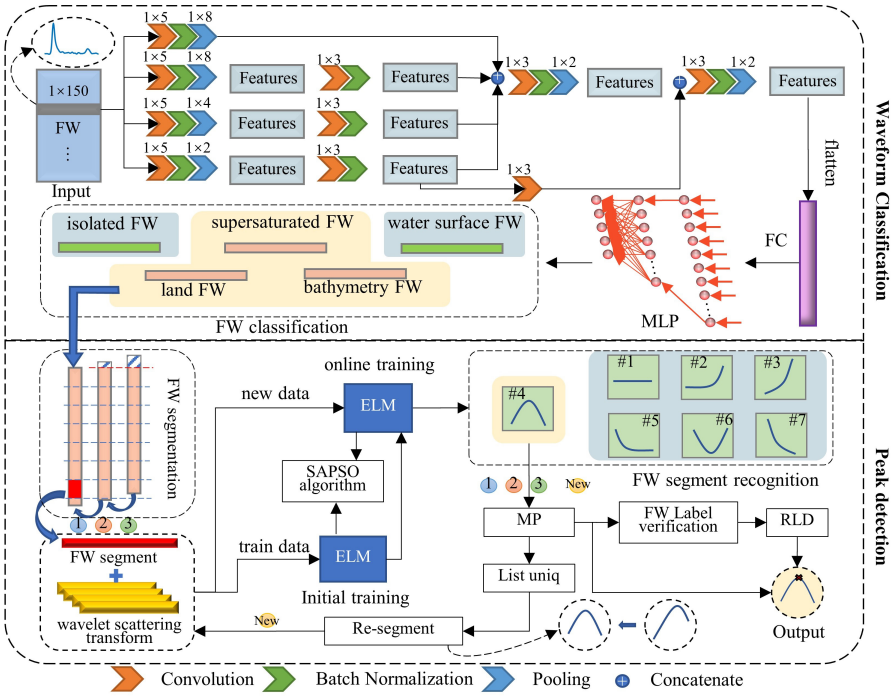


Fig. 1. Schematic of the waveform signal detection model.

2) Peak Detection: Ingeniously, the peak detection task is transformed into waveform segment identification with peaks and peak extraction on account of the newly constructed FW expression model. The waveform fragments are obtained through a leapfrog sliding window, and then the target waveform fragments with peaks are identified based on the proposed OS-SAPSO-ELM algorithm. Finally, peak extraction is performed on waveform segments that are identified as containing peaks.

B. Waveform-Type Determination

Based on the category waveform visualization analysis, it is found that there are several other types of waveforms that are significantly different from both the traditional water and land waveforms in morphology. Obviously, it is not appropriate to simply divide the waveform into water waveform and land waveform. Therefore, in this article, waveforms are divided into five categories based on similar morphological structures: isolated waveforms, supersaturated waveforms, land waveforms, water surface waveforms, and bathymetric waveforms, as shown in Fig. 2. The signal marked in Fig. 2 with ① is the isolated waveform which has no effective echo component and high morphological similarity. In the conversion between optical pulse and electronic signal, once the energy of the reflected signal exceeds the maximum threshold limited by the hardware, the peak of the obtained waveform will disappear, which is the named the supersaturated waveform corresponding to ② in Fig. 2. The waveform labeled ③ is the water surface waveform, which only involves the echo signal directly reflected from the water surface, without penetrating the water body. This type of waveform appears as a single peak, and the amplitude range is too large to distinguish. The land waveforms marked with ④ usually appear as a terrestrial return signal. Waveforms in vegetation coverage areas often show two or more echoes. However, areas of low-lying vegetation or bare ground appear as a single echo, which are related to signal energy attenuation. Consequently, the water surface waveform defined in this work is an echo with a relatively small amplitude, which is different from the land echo due to scattering and absorption of water. Bathymetric waveform marked with ⑤ is designed to describe the waveform that penetrates the entire water column and reaches the seabed, which is characterized with two effective peaks.

C. Waveform Classification

Most of the FW classification approaches focus on threshold setting or feature-based classifier construction. From the morphological perspective, FW can be regarded as sequential signals whose frequency and amplitude change with time, similar to sound, music, biomedical, and seismic signals. Universal classification models for sequential signals include the recurrent neural network (RNN) [43], long short-term memory (LSTM) [44], and CNN [45], [46]. The experiments have shown that the 1-D CNN model has a simple structure and a fast training and good ability to capture local information [47]. So a novel hierarchical context information fusion network is built up for FW classification based on CNN by combining feature maps and context information. Different from the conventional waveform classification models, the proposed approach requires neither excessive manual intervention nor complex feature extraction and compression processing.

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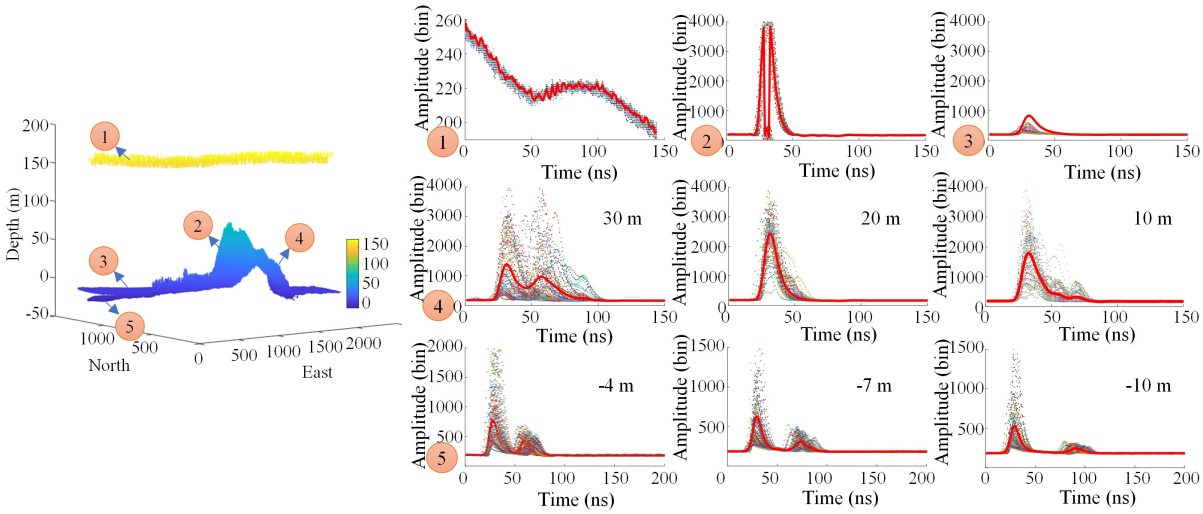


Fig. 2. Morphological structure of five types of waveforms. ① Isolated waveform. ② Supersaturated waveform. ③ Water surface waveform. ④ Land waveform. ⑤ Bathymetric waveform.

For the waveform signal $\mathcal{F} \in \mathcal{R}^{1\times 150}$, the convolution operation for waveform f_i with kernel g is defined as follows:

$$(\mathcal{F} * g)(x) = \sum_{x_i \in \mathcal{N}_x} g(x_i - x) f_i. \tag{1}$$

The contrast loss can be expressed as follows:

$$\text{Loss} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{i=1}^{T} y_i^n \log\left(p_i^n\right) \tag{2}$$

where y_i^n is the label of the nth waveform, and p_i^n is the probability that the nth waveform of the network output belongs to class i. Here, the rectified linear unit (ReLU) activation function is used, which is defined as follows:

$$\text{ReLU}(x) = \begin{cases} x, & \text{if } x > 0 \\ 0, & \text{if } x \le 0. \end{cases} \tag{3}$$

First, low-level feature information of the input is obtained by the convolution module, which includes the convolution layer, batch normalization, and ReLU activation function. Then, three different scales (i.e., 2, 4, and 8) of the pooling layer are used to reduce the dimensions of the feature map to collect multiscale feature representation. Second, for each size of feature map, two convolution layers and batch normalization are applied to obtain high-level information at a different scale; the output dimensions of which are set to 64. Third, the high-dimensional feature maps of the three scales are sampled uniformly to the same resolution using a different convolution kernel size. Furthermore, to retain the raw data distribution information, a convolution operation is applied to the input and sampled to the same resolution. Then, all the feature maps are concatenated and fused using the convolution layer to aggregate the multiscale information. Fourth, for the fusion information, the concatenate and convolution operations are used to aggregate the context information to facilitate the classification operation, and the final feature map is flattened. Finally, three fully connected layers and two dropout layers are used to integrate the features and avoid overfitting, for which the output channels are set to 1024, 256, and 5, keeping the probability of the dropout layer as 0.7. The output number of the final layer neurons is the class label for each input.

D. Peak Detection

1) Waveform Expression: Peak detection is a well-developed research subject and has achieved remarkable success. Waveform peak detection, similarly, as one of the most acceptable methods, is widely investigated. The achieved FW from the sensor is a superposition of environmental noise and single or multiple reflection signals of the target. For example, a bathymetric waveform can be expressed as the energy superposition of the water surface return, water scattering, seabed return, and noise [48]. It can be considered as a combination of M echoes, which can be described by a Gaussian function. Thus, the waveform can be expressed as the superposition of M Gaussian components

$$y = f(x) = \sum_{i=1}^{M} A_i e^{-(x-\mu_i)^2/2\omega_i^2} + N \tag{4}$$

where A_i, ω_i, and μ_i represent the parameters of the ith Gaussian component, that is, pulse amplitude, distance, and half width, respectively; and N represents the environmental noise. In a bathymetric waveform, noise is mainly generated by waves and backscattering. Therefore, seeking a higher precision approximation (optimization) model based on the iterative calculation of the fit waveform is considered as an arduous task.

To avoid complicated iterative parameter determination, and ensure accuracy and efficiency in signal detection, a new waveform representation theory is proposed. Here, the waveform signal is simply regarded as a combination of smooth signal, rising signal, and falling signal, marked out with #1, #2, and #3 in Fig. 3, respectively. Nine combinations can be formed through different permutation and combination. Among them,

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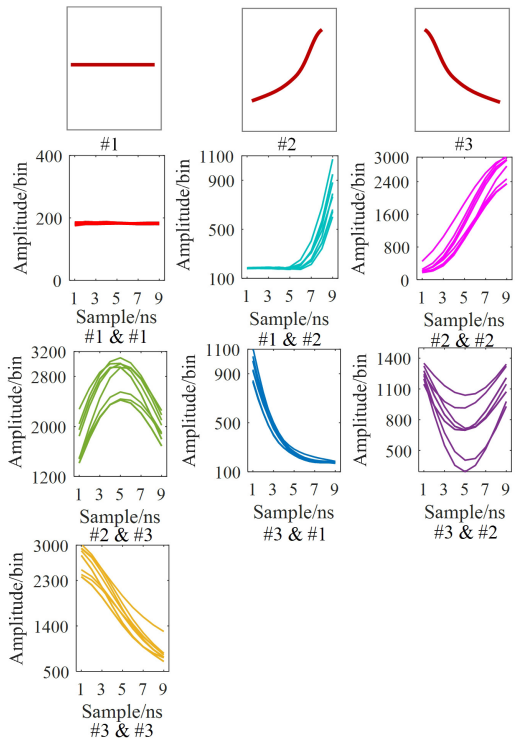


Fig. 3. Three basic components and the corresponding seven waveform segments.

the combinations of #2 & #1 and #1 & #3 are deliberately ignored due to nonexistence in practice. At this point, the signal detection task is transformed into waveform segment identification with peaks and peak extraction.

2) Waveform Segmentation: Waveform segmentation is required before performing waveform segment recognition. For waveform segmentation, a sliding window of length L, which depends on the specific system under consideration, is incorporated to divided the signal into segments with the same length, which are named waveform fragments. Among them, the peak only exists in the combination segments #2 and #3, which is considered as the target segment that requires special attention and further analysis. On the contrary, the remaining six types of segments are regarded as nontarget segments. It should be emphasized that the choice of window length is significant for waveform segment recognition. To avoid the classification error caused by the large deviation between the waveform fragment with peak and the standard target waveform fragment, a leapfrog sliding window waveform segmentation approach is designed. In addition to the original waveform being input, the waveform of a certain timescale ($1/3L$ and $2/3L$) is skipped and the remaining part is redivided, that is, the waveform undergoes multiple waveform segmentation and classification operations.

3) Waveform Fragment Classification: To realize the recognition of the target waveform segments, an online sequential extreme learning machine (OS-ELM) classification model improved by simulated annealing particle swarm optimization (SAPSO) is proposed. In comparison to the traditional signal detection methods, the application of artificial neural networks makes peak detection more flexible and intuitive [49], [50]. As a least-squares algorithm based on a single hidden layer feedforward neural network, extreme learning machine (ELM) can achieve the best performance, with the smallest training error and weight norm at the same time, and it is used to solve regression and classification issues [51].

For a standard ELM, it is assumed that there are N groups of arbitrary training samples (X_i, Y_i), where $X_i = [x_1, x_2, \ldots, x_n]^T$ is the input training sample with normalization processing and $Y_i = [y_1, y_2, \ldots, y_n]^T$ indicates the corresponding output. The output function of the hidden layer can be defined as follows:

$$o_j = \sum_{i=1}^{L} \beta_i g\left(\omega_i \cdot x_j + b_i\right) \tag{5}$$

where $\omega_{i,j}, b_i \in (-1, 1)$, and $j = 1, \ldots, N$; β_i represents the output weight; $g(x)$ is the excitation function; $\omega_i = [\omega_{i,1}, \omega_{i,2}, \ldots, \omega_{i,n}]^T$ denotes the input weight; and b_i is the threshold of the ith hidden layer unit. In this work, the feature matrix contains waveform segments and scattering transform extracted by the deep wavelet scattering network [52] to capture important information. The wavelet scattering network is a framework for automatic extraction of relevant compact features, which is also named the deep network. Each of its layers is usually composed of three parts—convolution, nonlinearity, and averaging—as illustrated in Fig. 4 (ψ is the wavelets, ϕ is the scaling function, and f is the input waveform segment). Among them, the convolution is performed by wavelets. The modulus operator used for filtering of the wavelet low-pass filter is similar to pooling.

Step1, first, a wavelet low-pass filter, the zeroth-layer scattering function, is used to average the input signal. After the averaging operation, high-frequency details in the signal will be completely removed.

Step2, by performing continuous wavelet transform on the signal to generate a set of scale coefficients, the details lost in the first step can be recaptured in subsequent layers. A nonlinear operator (modulus in this case) is applied to the scale coefficients. Then, a wavelet low-pass filter is used to filter the output to generate a set of first-order scattering coefficients.

Step3, the same process is repeated to obtain the second-layer scattering coefficients, that is, the scale coefficient output of the previous layer becoming the input of the operation of the subsequent layer. Then, we apply the same modulus operator and filter the output using a wavelet low-pass function to obtain the second-order scattering coefficients.

The scattering network contains three layers: the first layer performs averaging but loses details of the signal; the second layer recaptures the lost details, similar to the scale-invariant feature transformation function; the third layer provides supplementary information that can improve classification. These coefficients, which are collectively called the scattering features, are used as input for waveform fragment classification.

Actually, it is technically to obtain a universal model using a certain sample dataset. If the sample dataset is added to the

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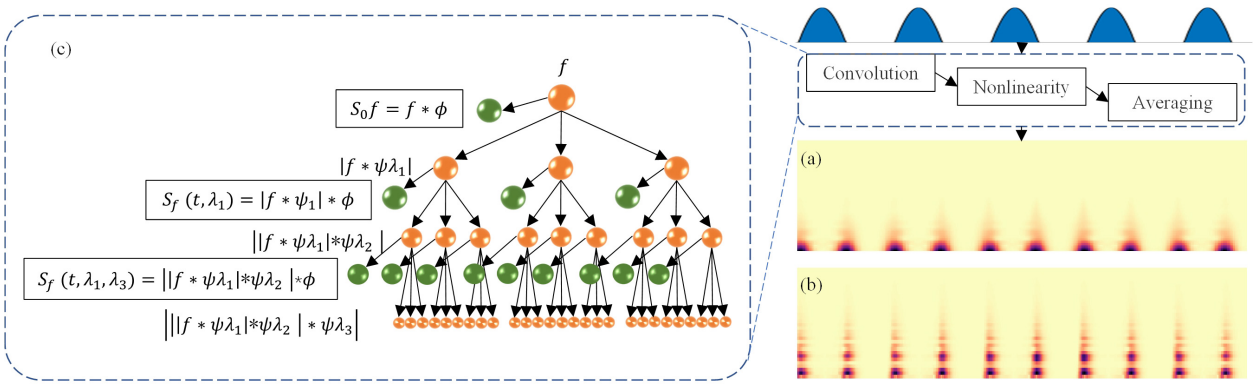


Fig. 4. Scattergram of scalogram coefficients. (a) First-order scattering coefficients. (b) Second-order scattering coefficients. (c) Tree view of the scattering transform.

training net, the batch learning algorithm needs to be retrained on the old data and the new data, which is time-consuming. Huang *et al.* presented an OS-ELM method that integrates the online sequential mechanism and discard the trained data to reduce space consumption to meet real-time requirements [53], [54], [55]. More attractively, the generality of the model continues to improve as the training data increase.

Then, the goal of the ELM algorithm is to minimize the difference between the output of the model and the actual theoretical output. The above can be expressed as follows:

$$H\beta = Y \tag{6}$$

$$H = \begin{bmatrix} (\omega_1 \cdot X_1 + b_1) & \cdots & g(\omega_L \cdot X_1 + b_L) \\ \vdots & \vdots & \vdots \\ g(\omega_1 \cdot X_N + b_1) & \cdots & g(\omega_L \cdot X_N + b_L) \end{bmatrix}_{N\times L} \tag{7}$$

$$\beta = \begin{bmatrix} \beta_1^T \\ \vdots \\ \beta_L^T \end{bmatrix}_{L\times m}, \quad \text{and } Y = \begin{bmatrix} y_1^T \\ \vdots \\ y_L^T \end{bmatrix}_{L\times m}. \tag{8}$$

The hidden layer parameters are randomly generated, and the output weight of the hidden layer is directly determined using the least-squares method

$$\hat{\beta} = H^* Y \tag{9}$$

$$H^* = \left(H^T H\right)^{-1} H^T \tag{10}$$

where H^* indicates the Moore–Penrose generalized inverse of the hidden layer output matrix H. This article introduces the online sequence learning process based on the initial ELM batch learning process and improves the generalization ability of the model by increasing the training samples.

However, the input weight matrix and hidden layer threshold of the ELM are randomly produced by the algorithm and usually require a large number of hidden layer neurons to achieve the desired prediction accuracy. It still cannot avoid the shortcomings of repeated adjustment of the parameters of the single feedforward neural network in the initialization phase. Therefore, the SAPSO algorithm, an optimization algorithm based on swarm intelligence [56] and thermodynamic system [57], is used to realize the optimal parameter selection of ELM, and its workflow is as follows:

Step 1, normalize the sample. Set the number of neurons and hidden nodes, and select the kernel function sig.

Step 2, model parameter initialization. The parameters include the initial population size M, the initial temperature T_0, the cooling speed α, the speed v and position x of all the particles in the target space, the learning factors c_1 and c_2, the inertia factor $\omega \in [\omega_{\min}, \omega_{\max}]$, and the iteration termination condition which includes maximum iterations ($K_{\max} = 1000$) and minimum error standard (0.0001)

Step 3, perform simulation training on ELM according to (6)–(10), and calculate the fitness function according to the mean square error of the training sample error

$$E = \frac{1}{N}\sum_{i=1}^{N}\sqrt{(o_i - y_i)^2} \tag{11}$$

$$\Delta E = f^{k+1}(x_i) - f^k(x_i). \tag{12}$$

Step 4, iterative calculation. If ΔE is less than 0, accept x as the new current solution, and update the system temperature according to (13); otherwise, the temperature remains unchanged

$$\exp\left(-\frac{\Delta E}{T}\right) > \text{rand}, \quad \Delta E < 0 \tag{13}$$

$$T^{k+1} = \alpha T^k \tag{14}$$

where rand is a random number between (0, 1), $f(x_i)$ is the fitness value of the ith particle, and T is the temperature under the current iteration.

Step 5, according to (15) and (16), update v and x of the $(k+1)$ th iteration, and adjust ω adaptively according to (17). Then, find the current optimal position p_{id} and global optimal position of all the particles g_{id}

$$v_{id}^{k+1} = \omega v_{id}^k + c_1 r_1\left(p_{id}^k - x_{id}^k\right) + c_2 r_2\left(g_{id}^k - x_{id}^k\right) \tag{15}$$

$$x_{id}^{k+1} = x_{id}^k + v_{id}^{k+1} \tag{16}$$

$$\omega = \omega_{\max} - \frac{\omega_{\max} - \omega_{\min}}{K_{\max}}. \tag{17}$$

Step 6, when the system meets the iteration termination condition, the iteration is terminated. g_{id} is the optimal (ω, b). β and H can be calculated using (6)–(9); otherwise, return to Step 3.

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Finally, the hidden layer output weight β_0 and the hidden layer output matrix H_0 in the initial network can be determined as follows:

$$\beta_0 = K_0^{-1} H_0^T Y_0 \tag{18}$$

where $K_0 = H_0^T H_0$. When a training sample is added, the hidden layer output matrix H_{t+1} and the output weight β_{t+1} are

$$H_{t+1} = \begin{bmatrix} g\left(\omega_1^T \cdot X_1^{(t+1)} + b_1\right) & \cdots & g\left(\omega_L^T \cdot X_1^{(t+1)} + b_L\right) \\ \vdots & \vdots & \vdots \\ g\left(\omega_1^T \cdot X_{N_{t+1}}^{(t+1)} + b_1\right) & \cdots & g\left(\omega_L^T \cdot X_{N_{t+1}}^{(t+1)} + b_L\right) \end{bmatrix}_{N \times L} \tag{19}$$

$$\beta_{t+1} = \beta_t + K_{t+1}^{-1} H_{t+1}\left(Y_{t+1}^T - H_{t+1}^T \beta_t\right) \tag{20}$$

where, $K_{t+1}^{-1} = K_t^{-1} - K_t^{-1} H_{t+1} H_{t+1}^T K_t^{-1} (I + H_{t+1}^T K_t^{-1} H_{t+1})^{-1}$.

It can be seen from the complete OS-SAPSO-ELM algorithm workflow that the optimal search of ω_i and b_i by the SAPSO algorithm improves the robustness of ELM. Therefore, ELM is able to achieve better prediction effect with a small number of hidden layer neurons. Moreover, OS-SAPSO-ELM overcomes the shortcomings of overlapping local extreme points in previous algorithm, enhances network robustness, and improves the convergence speed of the subsequent evolutionary algorithm and the generalization of the network.

4) Signal Detection: Once the waveform segment containing peak is identified accurately, the difficulty of peak detection is remarkably reduced, which can be easily completed using the MP algorithm with high accuracy, as shown in Fig. 5. Since complete FW is needed to be repeatedly input three times, it is necessary to remove the repeated peaks. Subsequently, the waveform segment is reintercepted with the peak as the center and L as the length, further as the input to perform peak detection. The significance of this step is to verify whether the extracted peak is correct or not. However, for a bathymetric waveform, especially one from extremely shallow water, it is arduous to detect the bottom echo signal correctly owing to substantial mixing of the water surface and bottom reflection signals. Therefore, once the detection peak of the bathymetric waveform is not equal to 2, the RLD peak detection method will be enabled. This makes the model suitable for signal detection in different scenarios with variable depths.

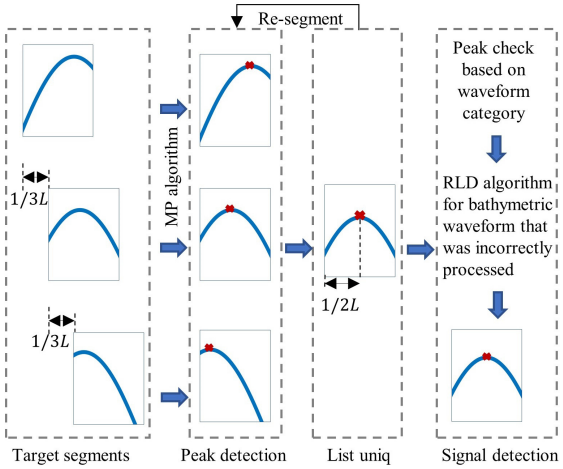


Fig. 5. Work flowchart of peak detection.

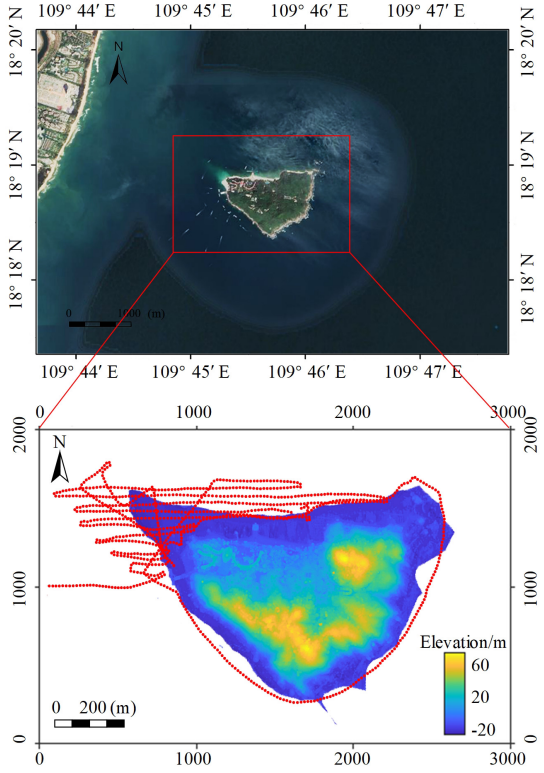


Fig. 6. Schematic of the study area.

III. Results

A. Study Area and Datasets

Wuzhizhou Island is located in Haitang Bay to the northeast of the city of Sanya in Hainan Province, China, lying between Nanwan Monkey Island and Yalong Bay (Fig. 6). The island, which lies 2.7 km off the coast, is 1400×1100 m (E − W × N − S) and covers an area of 1.48 km^2. An ALB survey was conducted on January 20, 2013, by the First Institute of Oceanography, MNR, using an Aquarius system launched by Optech, which is an innovative compact solution for mapping coastal and inland waterways. The nominal operating altitude of the aircraft during the survey was 300 m with a constant laser view angle of 20°. The position and elevation of the aircraft were determined using a differential global positioning system with real-time kinematic and postprocessed kinematic solutions and the inertial measurement unit. In addition, the water depth data in the shallow water area of the experimental area were collected by the ship-borne single-beam sounding system in 2013. Its position is marked with red dots in the local enlarged image in Fig. 6.

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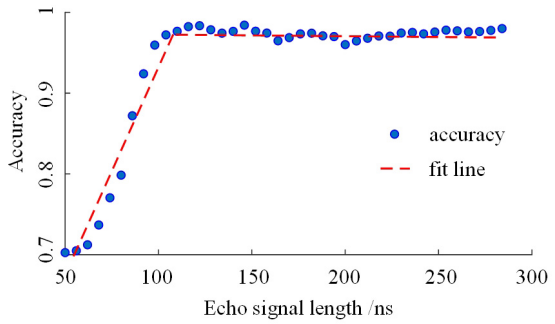


Fig. 7. Relationship between waveform length and classification accuracy.

B. Determination of Input Waveform Length

To obtain the optimal waveform length and compress data amount, 50 000 samples (10 000 samples per type) are selected for experimentation. Waveforms of different lengths (i.e., an arithmetic sequence covering the range of 50–285 ns with a 5-ns interval) were input into the classifier. The classification accuracy statistics corresponding to different inputs are shown in Fig. 7. From the results, it is notable that when the waveform length is larger than 120 ns, the variation in the classification accuracy tends to be stable (97%), which indicates that the effective echo record within this range is relatively complete. Thus, for further analysis, the waveform length of the input classifier was set to 150 ns. This operation substantially reduces the amount of data, without scarifying the accuracy and flexibility on classification.

C. Waveform Classification and Accuracy Assessment

In the demonstration, 60 000 waveform files were selected for each sample type. The preparation of the samples integrated various information such as the registered satellite image (Worldview-2), thresholds, and waveform morphology parameters based on waveform decomposition [60]. According to the waveform amplitude range from 180 to 300 bin and echo component parameters (there is no effective echo component, the number of peaks and the corresponding morphological parameters are nan values), the isolated waveform can be selected. The supersaturated waveform can be identified by threshold (maximum value more than 3400 bins) and waveform shape parameters (two or more effective echoes, large effective echo kurtosis, maximum amplitude deviation of adjacent two points greater than 3000 bins, etc.). The terrestrial and nonterrestrial waveforms were divided via registered satellite images. Waveform morphology parameters (the number of effective echo, the first echo amplitude, cross-sectional area, etc.) were used for the division of water surface waveform and effective bathymetry waveform. The files of each sample type were randomly divided into two groups: training data and test data. The main parameters of the training model are set as follows: learning rate is 0.001, batch size is 3000, iteration is 50, and epoch is 30. To evaluate the performance of the FWSD model on waveform classification, waveform-characteristic-based random forest (RF) approach, RNN, and LSTM method are also conducted for comparison. The waveform was decomposed using the Levenberg–Marquardt algorithm

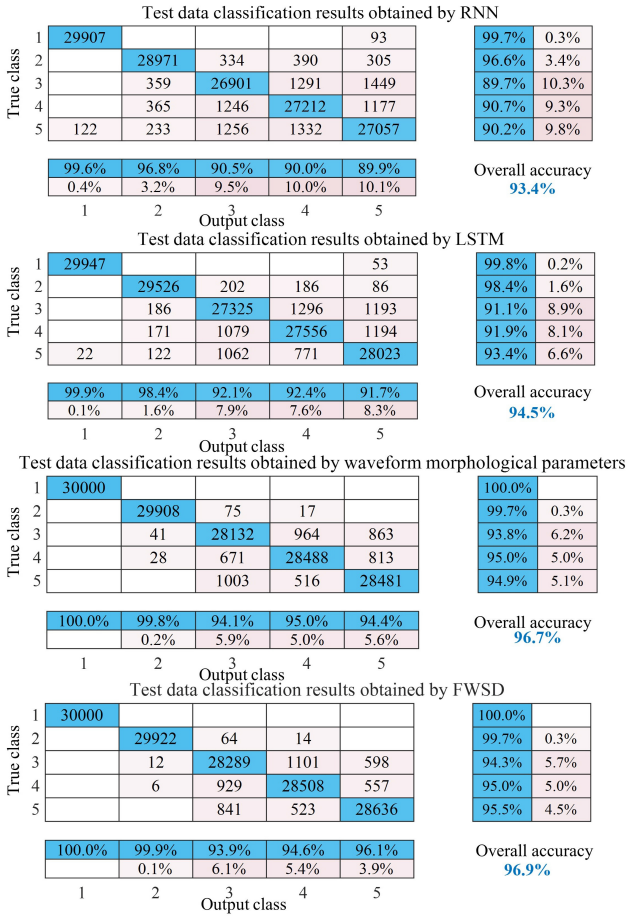


Fig. 8. Classification accuracy consumption for four classification methods.

optimized generalized Gaussian model (LM-GGM), and 15-D features (the number of peaks, amplitude of the first echo, second echo and last echo peak, full-width at half-maximum of the first echo, mean of the first echo, standard deviation of the first echo, kurtosis of the first echo, root mean square of the first echo, form factor of the first echo, peak factor of the first echo, impulse factor of the first echo, margin of the first echo, skewness of the first echo, and pulse area of the first echo) were extracted for feature matrix [58], [59]. As presented in Fig. 8, the classification accuracy of the FWSD model is better than that of other methods through comparative experiment of test samples. The labels ranging from 1 to 5 correspond to isolated waveform, supersaturated waveform, land waveform, water surface waveform, and bathymetric waveform, respectively. Due to feature extraction, the RF classification model based on waveform parameters takes 17 min 34 s to classify test data, which is much longer than the other three deep learning models (less than 1 min). The combination of local descriptors and context information makes the classification ability of the FWSD model better than the traditional RNN and LSTM.

To visually show the waveform classification effect of the FWSD model, three rectangular areas in the test area with coverage of water and land were selected for demonstration. As presented in Fig. 9, the waveform recognition results of the three regions are all satisfactory and meet our expectations.

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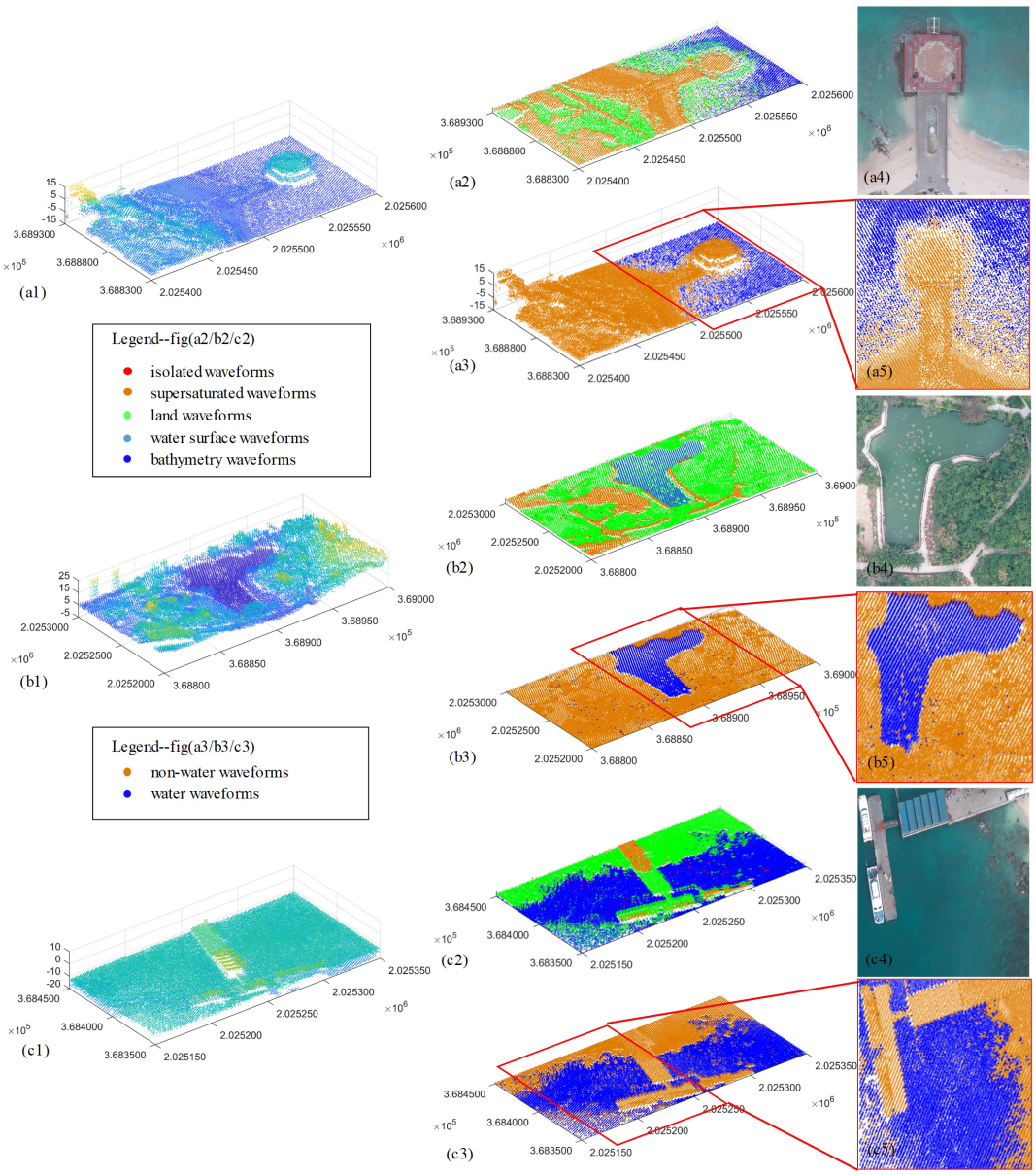


Fig. 9. Waveform classification results for three areas. (a)–(c) Area numbers. #1 is the depth, #2 is the waveform classification result (including five categories), and #3 is the water–land classification obtained based on #2. Subpictures #4 and #5 are the corresponding airborne red-green-blue (RGB) images and category diagrams in the red box, which are selected for detailed display.

Further inspection reveals that the signals received from bare surfaces (e.g., roads, plank roads, and ships) are mostly supersaturated waveforms. In (b2), the aquatic plants in ponds and peripheral plants are fully distinguished as expected. Inevitably, the signals in areas (a) and (c) are mixed with strong water surface reflection signals recognized as land waveforms. As aforementioned, water surface waveforms include not only the fourth type of waveform but also some echo signals with larger amplitude. However, once the bathymetric waveform is identified as well, these signals can be eliminated through coverage determination.

D. Window Length Determination

After water and land waveforms are successfully separated, signal detection is performed immediately. As mentioned in Section II-D2 waveform segmentation, the setting of the L value is very important. The raw waveform was sequentially sent into the OS-SAPSO-ELM algorithm through leapfrog sliding window for waveform segment identification. Among them, those identified as target waveform segments will be performed peak extraction. At this point, peak extraction becomes relatively simple. In other words, it is a pretty simple case of deriving MP. The next problem to be considered is to appropriately set the window length L. A series of preliminary experiments were conducted, and the results are shown in Fig. 10. It can be seen from the results in Fig. 10 that if L is too small, a large number of waveform fragments will be obtained, in which feature records are insufficient and it poses a challenge to the computational load. Conversely, if L is too large, each waveform segment will contain more than two trends, especially for the second (or greater) echo signals, thereby reducing the recognition accuracy of the

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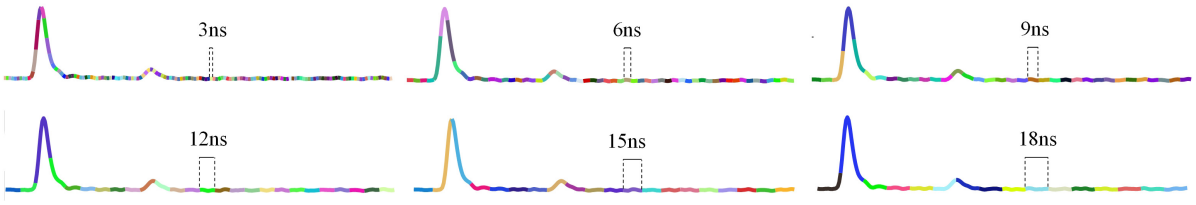


Fig. 10. Waveform segmentation results based on different window lengths L.

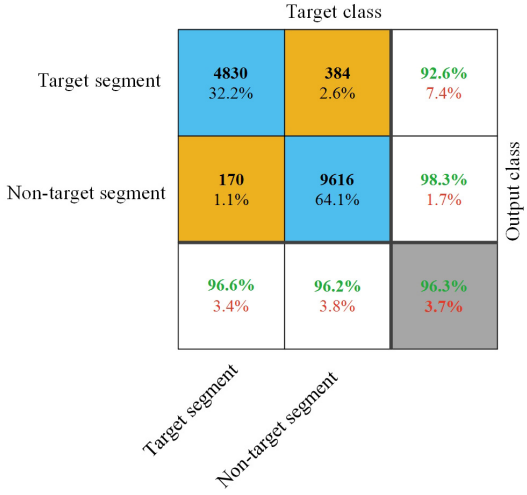


Fig. 11. Classification accuracy of waveform segments.

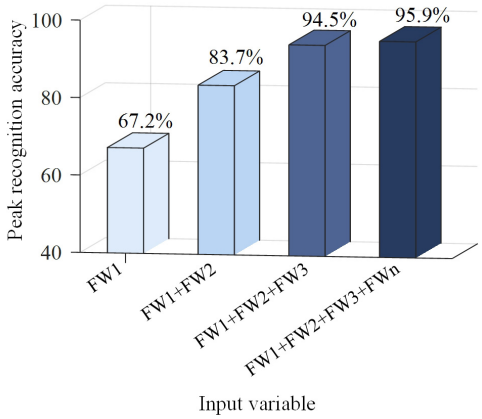


Fig. 12. Peak recognition accuracy corresponding to different inputs.

waveform segment. Here, a value of L of 9 ns was selected to ensure preservation of the waveform characteristic and a reasonable number of fragments.

E. Peak Detection and Accuracy Assessment

The first-order derivative of the filtered waveform is performed to determine the amplitude variation trend, and then different types of waveform fragments are segmented and normalized. Among them, the waveform fragment labeled class 4 uses the MP method to detect the peak. For each category, 30 000 samples (10 000 for class 4 fragments, 20 000 for other types of fragments) were selected. Half of them were used for training and the remaining were used for testing. The confusion matrix calculated by the FWSD model, with the characteristic matrix composed of the amplitude vector of the waveform fragment and the corresponding wavelet scattering transformation as the input, is shown in Fig. 11. A high recognition accuracy of 96.3% is achieved. Inevitably, certain waveform segments containing peaks are morphologically different from the standard fourth-type waveform segments. For further improvement of recognition accuracy and confirm the misrecognition components, the complete waveform based on leapfrog sliding window is segmented three times, and the corresponding fragments FW1, FW2, and FW3 are obtained. Then, remove duplicate peaks. Subsequently, the resegment (FWn) with the retained peak as the center and window length L is input into the classifier. Theoretically, FWn is the closest to the fourth standard waveform morphologically, and it can accurately determine whether peak extraction is correct or not. The point cloud coordinates can be calculated by combining the extracted peak with the waveform category and attitude parameters. Experiments on 3000 waveforms were conducted, and peak extraction accuracy corresponding to different inputs is shown in Fig. 12. The results show that the peak identification accuracy is significantly improved with the input of FW2 and FW3 (16.5%–27.3%). Similarly, misidentified peaks are effectively eliminated, and peak extraction accuracy is also improved with FWn input.

To qualitatively and quantitatively analyze the signal detection results, a signal extraction comparison experiment was carried out in two rectangular areas of the experimental area based on the LM-GGM algorithm, RLD algorithm, and FWSD model. As shown in Fig. 13, area (a) is located underwater with underwater reefs. In comparison, area (b) is relatively rich in features that include underwater topography, exposed reefs, beach, and buildings. Fig. 13 (a1 & b1) shows the depth obtained by the acquisition system software. Judging from Fig. 13 (a2 & b2) and (a3 & b3), we can see that most of the nonbathymetry and invalid waveforms are eliminated after waveform classification. The signal detection results based on LM-GGM, RLD algorithm, and FWSD can more suitably describe the actual topography as shown in Fig. 13 (a4 & b4), (a5 & b5), and (a6 & b6), respectively. To demonstrate the performance of three different signal detection models, six circular areas with a diameter of 30 m, marked as ①–⑥ in Fig. 13, were selected and subjected to

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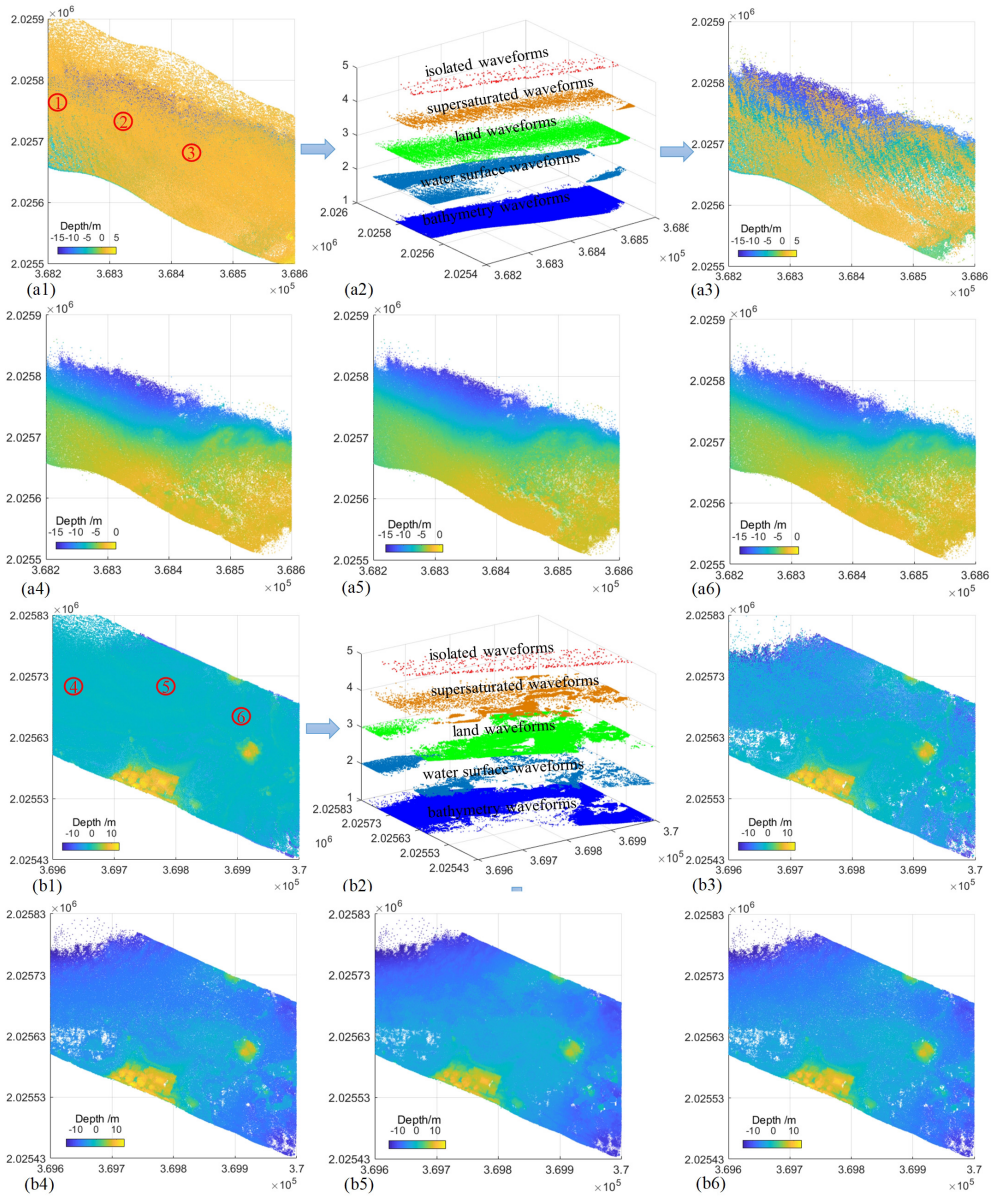


Fig. 13. Water depth maps obtained from the three algorithms. (a) and (b) Areas of interest, #1 depth calculated using the acquisition system software, #2 FW classification result, #3 depth after excluding invalid waveforms, and #4, #5, and #6 depth obtained using the LM-GGM model, RLD algorithm, and the FWSD model, respectively.

TABLE I

COMPARISON OF ACCURACY AND TIME CONSUMPTION OF THE THREE METHODS

Arguments		Raw data	LM-GGM	RLD	FWSD
Time loss			28 min 37 s	19 min 22 s	9 min 15 s
RMSE	1	8.729 m	0.148 m	0.134 m	0.131 m
	2	7.471 m	0.143 m	0.148 m	0.147 m
	3	6.762 m	0.145 m	0.135 m	0.131 m
	4	3.181 m	0.153 m	0.130 m	0.132 m
	5	2.290 m	0.161 m	0.134 m	0.134 m
	6	2.138 m	0.172 m	0.142 m	0.145 m

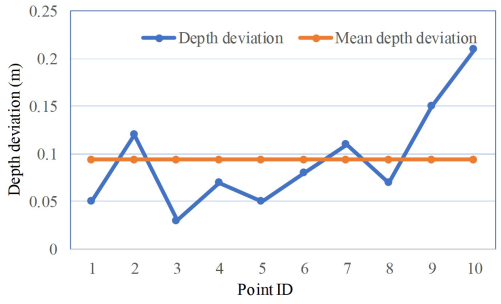


Fig. 14. Water depth deviation statistics between single-beam echosounder and ALB based on the FWSD model.

statistical analysis, involving time-consuming and accuracy, as shown in Table I. In addition, the depth obtained by the FWSD method were compared with the single-beam bathymetric data collected during the same period after tide correction and reference correction (shown in Fig. 6), as shown in Fig. 14. The average deviation of water depth was 0.09 m. It can be seen from Table I that the FWSD model can be perfectly adapted to different water depths and obtain satisfactory

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waveform signal detection output on both time consumption and recognition accuracy.

IV. Conclusion

The single-wavelength ALB systems are widely used to determine DEMs of the terrestrial surface and the seafloor owing to their superiorities on topographic survey and bathymetry. However, the limitations of the current equipment indicate that waveform mixing is still a technically challenging issue, which makes arduous to identify the waveform type and calculate the point cloud position. Facing this issue, this article proposes a simplified FW expression model and used the machine learning methods to process single-wavelength ALB waveforms, making the waveform processing accurate, intuitive, and efficient.

It is difficult to get accurate detection results without considering waveform category. The inevitable waveform decomposition and feature parameter extraction make the waveform-feature-based waveform classification characterized as a highly time-consuming and intensive computational cost approach. The FWSD model proposed in this article incorporates a CNN with the capability to comprehensively use the local, global, and context information of FW to achieve waveform classification and provide constraints for the next signal detection. The FW contains five types of recognized characteristics: isolated waveforms, land waveforms, supersaturated waveforms, water surface waveforms, and bathymetric waveforms. Compared with the traditional waveform classification approach, the proposed method demonstrates better classification accuracy and efficiency and does not require empirical threshold settings. Moreover, another attractive feature of the method is that once the network is well-trained, it can realize real-time waveform classification without further intervention. The waveform classification process eliminates isolated waveforms and water surface waveforms, remarkably reducing the workload on peak detection.

Based on the comprehensible FW theoretical model, complex signal detection tasks are skillfully transformed into two subtasks: identification of waveform segments that have peaks and peak detection. Leapfrog sliding-window-based waveform segmentation is designed to execute multiple split commands on the waveform, which proved to be necessary to improve the recognition accuracy of the target waveform segment. In waveform segment recognition, an improved OS-SAPSO-ELM algorithm combined with the wavelet scattering network is proposed. Among them, rich feature matrix, online learning mechanism, and parameter optimization improve the generalization ability and classification performance of ELM. In addition, the detected waveform segments containing peaks are resegmented and input into the waveform segment classification model. This operation can perform a more accurate secondary screening of the peaks. In comparison to the LM-GGM algorithm, the RLD algorithm, and the solution obtained using the instrument software, the proposed model achieved more satisfactory detection efficiency and accuracy. The Gaussian decomposition model is more sensitive to the shape of the waveform. When the echo waveform shape deviates greatly from the Gaussian waveform, the possibility of incorrect decomposition of the waveform becomes greater. Moreover, the position of the peak will be shifted owing to improper filtering and Gaussian decomposition. Although the RLD algorithm is sensitive to small peaks, it is easy to introduce ineffective target points, such as suspended matter and environmental noise. The FWSD model is insensitive to error, which reduces the extraction of false peaks to a certain extent, but an obvious disadvantage is that the peak position is affected by sampling frequency. By comparing with the single-beam measured water depth data, it is proven that the signal detection results of the new model can meet the engineering accuracy requirements.

Correction for the sea surface angle of incidence is vital for seabed coordinate calculation, that is, accurate calculation of the topography of the water surface is crucial to the correction of the incident angle of the beam. However, without sufficient water surface topography data, accurate incidence angle calculations are not achievable. In this article, waveforms shot into areas of seawater coverage are deleted, except the bathymetric waveform records the bottom echo. The deleted waveforms record surface echo signals, which could be used to calculate water surface topography. Therefore, in future research, we need to recall and make full use of these waveform data.

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